Algebra & Equations

ANSWER KEY

Section 1: Quadratics & Complex Numbers

1 What is the value of the discriminant for the quadratic equation $2x^2 - 4x + 5 = 0$?

a. -24 ✓

b. 24

c. -56

d. 56

Discriminant $\Delta = b^2 - 4ac = (-4)^2 - 4(2)(5) = 16 - 40 = -24$.

2 Simplify the complex number expression: i^{46}.

a. -1 ✓

b. 1

c. i

d. $-i$

Powers of i repeat every 4: divide $46 \div 4 = 11$ with remainder 2. Thus $i^{46} = i^2 = -1$.

3 What are the solutions of the equation $x^2 + 16 = 0$ over the complex numbers?

a. $\pm 4i$ ✓

b. ± 4

c. $\pm 2i$

d. $\pm 16i$

$x^2 = -16 \Rightarrow x = \pm\sqrt{-16} = \pm 4i$.

Section 2: Matrices & Determinants

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4 Find the determinant of the 2×2 matrix $A = [[4, 3], [2, 5]]$.

a. 14 ✓

b. 26

c. 10

d. 18

$\det(A) = (4)(5) - (3)(2) = 20 - 6 = 14$.

5 For what value of k does the matrix $M = [[k, 6], [2, 3]]$ have NO inverse?

a. 4 ✓

b. 6

c. 2

d. 3

A matrix has no inverse when its determinant is zero:
$(k)(3) - (6)(2) = 0 \Rightarrow 3k - 12 = 0 \Rightarrow 3k = 12 \Rightarrow k = 4$.

6 If $A = [[1, 2], [3, 4]]$ and $B = [[2, 0], [1, 3]]$, what is the element in row 1, column 2 of $A + 2B$?

a. 2 ✓

b. 4

c. 6

d. 8

Row 1, column 2 entry: $A_{12} + 2B_{12} = 2 + 2(0) = 2$.

Section 3: Polynomials & Remainder Theorem

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7 What is the remainder when $P(x) = 2x^3 - 4x^2 + 5x - 3$ is divided by $(x - 2)$?

a. 7 ✓

b. 5

c. 9

d. 3

By the Remainder Theorem, evaluate $P(2)$: $P(2) = 2(2^3) - 4(2^2) + 5(2) - 3 = 16 - 16 + 10 - 3 = 7$.

8 If $(x - 3)$ is a factor of $x^3 - kx^2 + 2x - 6$, what is the value of k?

a. 3 ✓

b. 2

c. 4

d. 5

Since $(x - 3)$ is a factor, $P(3) = 0$:
$3^3 - k(3^2) + 2(3) - 6 = 0 \Rightarrow 27 - 9k + 6 - 6 = 0 \Rightarrow 27 - 9k = 0 \Rightarrow k = 3$.

9 What is the degree of the polynomial $f(x) = (3x^2 - 1)^2(x^3 + 4)$?

a. 7 ✓

b. 6

c. 8

d. 12

Degree of $(3x^2 - 1)^2 = 2 \times 2 = 4$. Degree of $(x^3 + 4) = 3$. Total degree $= 4 + 3 = 7$.

Section 4: Systems of Equations & Inequalities

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10 What is the value of $x + y$ for the system: $3x + 2y = 16$ and $2x - 2y = 4$?

a. 6 ✓

b. 5

c. 7

d. 8

Add both equations: $5x = 20 \Rightarrow x = 4$. Substitute into $2(4) - 2y = 4 \Rightarrow 8 - 2y = 4 \Rightarrow 2y = 4 \Rightarrow y = 2$. Then $x + y = 4 + 2 = 6$.

11 Which point lies in the solution set of the system: $y > 2x - 1$ and $y \leq -x + 4$?

a. $(1, 2)$ ✓

b. $(2, 5)$

c. $(0, -2)$

d. $(3, 0)$

Test $(1, 2)$: $2 > 2(1) - 1 \Rightarrow 2 > 1$ (True), and $2 \leq -1 + 4 = 3$ (True). Thus $(1, 2)$ satisfies both inequalities.

12 If $\frac{x-3}{x+2} = 0$, what is the real solution for x?

a. 3 ✓

b. -2

c. 0

d. 5

A fraction equals zero when its numerator is zero and denominator is non-zero: $x - 3 = 0 \Rightarrow x = 3$ (with $3 + 2 = 5 \neq 0$).

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